



AI is not convinced AI is a major disinflationary force

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Introduction

In a recent note, we discussed how AI assesses the potential negative effects of AI on the unemployment rate (see “Will AI drive unemployment higher this year? AI thinks not”). While several AI tools agree that the impact is likely to be limited over the next year -- all three tools we referenced assign a 5% probability AI will lift the unemployment rate by 0.5pp or more over the next 12 months -- there was disagreement further out. The three tools assigned a 10-30% probability of more substantial disruption at the 5-year horizon.

In this note we take the same approach to assess the impact of AI on inflation. In contrast to the unemployment rate, where there appears to be considerable uncertainty about the size of the impact, our sense from client conversations is that there is a strong consensus that AI represents a substantial disinflationary force that will also exert meaningful downward pressure on interest rates over the coming years.

Does AI agree with this consensus? Surprisingly not.

Why AI is expected to be a significant disinflationary force

The case for AI to exert a meaningful drag on inflation is reasonably straightforward. AI is expected to substitute cheap capital / technology for more-expensive labor, driving input costs lower. By leading to sustainably higher productivity growth, it will push unit labor costs substantially lower, weighing on price pressures. To the extent that AI raises the unemployment rate, it will also reduce inflation through increasing labor-market slack, reducing wage growth, and weakening aggregate demand. Finally, AI represents an important equalizing force for entrepreneurs, ushering in countless small businesses with lower barriers to entry that will compete and drive margins lower. This is our understanding of the consensus view on why inflation and rates will be sustainably lower due to AI.

But there are countervailing forces to this narrative. In the near term at least, AI has triggered a substantial positive aggregate demand shock, as investment has surged in anything AI-related, including software, chips, computers and peripheral equipment, and data centers. These investments are competing with alternative demands for resources. In addition, AI is likely to represent an ongoing source of demand that could create bottlenecks and price pressures in key points of the supply chain -- the spillover to higher electricity prices is most obvious. Finally, while the initial impact of AI-driven higher unemployment would be weaker demand and softer inflation pressures, we expect policy levers would be triggered to ultimately limit the rise in unemployment and put a floor under aggregate demand. Fiscal policy is best placed to serve this redistribution role -- indeed stronger productivity implies faster GDP growth which must also imply stronger aggregate income growth, making redistribution possible -- but monetary policy could also be part of the initial response.

Authors

Matthew Luzzetti, Ph.D.
Chief US Economist

Brett Ryan
Senior US Economist

Justin Weidner
Economist

Amy Yang
Economist



So, while we find the disinflationary consensus to be compelling, there are forces that could at least limit the magnitude of these forces. What does AI say?

What dbLumina says

We begin by asking DB's proprietary AI tool, dbLumina, how it assesses the probabilities around inflation effects from AI over two different horizons -- the next 12 months and five years ahead. We use the following prompt:

I'm interested in studying the different probabilities of outcomes for AI's impact on US inflation. I have in mind: the probabilities it raises inflation by 0.1pp or more, has minimal impact on inflation, slightly reduces inflation and meaningfully reduces inflation. Please consider two different horizons: 1 year and 5 years. Define slightly reduces as -0.1pp to -0.4pp and meaningfully reduces as -0.5pp and more at both horizons. Please summarize the results in a table. Provide the reasoning for your findings as well as citations to academic papers in the table.

Over the next year, dbLumina sees risks as tilted slightly towards higher inflation from AI. There are roughly equal probabilities assigned to the outcomes where AI has a minimal impact on inflation (35%) and lifts inflation by any amount (40%). In contrast, AI lowering inflation over the next year has a total probability of only 25% (5% assigned to AI lowering inflation by 0.5pp or more over the next 12 months and 20% probability of a drag of -0.1pp to -0.4pp). dbLumina's rationale is outlined within the table in the final column. As we noted in the prior section, dbLumina cites the positive impact on aggregate demand through AI-related investment as well as a potential boost to energy prices as rationalizing risks of higher inflation.

Looking further out, the risks are better balanced. Five years ahead, dbLumina now sees a roughly 50% probability that AI lowers inflation by any amount; however, the overall distribution is roughly flat. While the modal outcome is a slight reduction in inflation (-0.1 to -0.4pp), dbLumina assesses it as more likely that AI lifts inflation by any amount over the five-year horizon (25%) than meaningfully reduces inflation over that period (20%). That skew does not fit the consensus view on AI's disinflationary impact.



Figure 1: Probability of AI's net impact on US inflation

Scenario (Net Impact on Inflation)	Quantitative Definition (Both Horizons)	Probability (Next 12 Months)	Probability (Next 5 Years)	Reasoning and Citations
1. Raises Inflation (Demand-side effects, investment boom, or bottlenecks dominate)	+0.1 pp or more	40%	25%	The initial phase of AI adoption is likely to be characterized by strong investment and demand-side pressures, potentially leading to a modest increase in inflation. The International Monetary Fund (IMF) also projects a near-term rise in inflation due to higher investment and aggregate demand . Over a longer horizon, the demand-pull effects of sustained AI-driven growth could still lead to higher inflation, as suggested by some long-term scenarios in a BIS working paper where demand effects eventually dominate.
2. Minimal Impact (Supply and demand effects largely offset; slow adoption)	-0.09 pp to +0.09 pp	35%	25%	The impact of AI on inflation in the short term could be balanced, with productivity gains partially offsetting increased demand. The precise timing and magnitude of these opposing forces are uncertain, leading to a significant probability of a minimal net effect . The long-term impact of AI on inflation remains highly uncertain, with competing forces of productivity gains and increased demand potentially neutralizing each other.
3. Slightly Reduces Inflation (Supply-side efficiency gains, cost reductions, or weak demand)	-0.1 pp to -0.4 pp	20%	30%	Early productivity gains from AI could begin to put downward pressure on prices, but this effect is expected to be limited in the first year as adoption is still in its early stages. As AI becomes more widely adopted, its productivity-enhancing and cost-reducing effects are expected to become more significant, leading to a moderate reduction in inflation.
4. Meaningfully Reduces Inflation (Widespread cost-cutting, extreme productivity boom, or demand shortfall)	-0.5 pp or less	5%	20%	A significant reduction in inflation within a year is considered unlikely as the widespread adoption and deep integration of AI required to achieve such an outcome takes time.

Source: Deutsche Bank Research

This result is somewhat surprising, as it goes against the strong consensus that AI is likely to represent a transformative disinflationary force. Given the tension, it is important to investigate whether any difference in assessments is unique to dbLumina or is represented across a broader selection of AI tools.

What do other AIs say?

To test the robustness of our results in the previous section, we provide the same prompt to ChatGPT 5.2 and Claude Opus 4.6. Though there are some differences in probabilities, the main point of tension found in dbLumina's analysis — that the risks to inflation are likely to be more balanced than commonly assumed — is also present across other AI tools.

For example, the alternative AI tools view the most likely outcome as minimal impact over the next year (50-60% probability across AI tools). All tools see a meaningful reduction in inflation at the one-year horizon as a tail risk (5%) and see increases in inflation as more likely than a meaningful reduction. dbLumina is the most skewed at 40% vs. 5%, but ChatGPT (20% vs. 5%) and Claude (25% vs. 5%) agree with the direction of the skew. Across tools, the role of the positive aggregate demand shock from AI-related investment features consistently, as does the potential boost to electricity prices.



Figure 2: Summary of AI views on the inflation effects of AI

1-year horizon				
	Increases Inflation (+0.1 pp or more)	Minimal impact (-0.09 pp to +0.09 pp)	Slightly reduces (-0.1 pp to -0.4 pp)	Meaningfully reduces (-0.5 pp or less)
dbLumina	40%	35%	20%	5%
ChatGPT 5.4	20%	60%	15%	5%
Claude Opus 4.6	25%	50%	20%	5%
Average	28%	48%	18%	5%
5-year horizon				
	Increases Inflation (+0.1 pp or more)	Minimal impact (-0.09 pp to +0.09 pp)	Slightly reduces (-0.1 pp to -0.4 pp)	Meaningfully reduces (-0.5 pp or less)
dbLumina	25%	25%	30%	20%
ChatGPT 5.4	15%	25%	40%	20%
Claude Opus 4.6	20%	25%	35%	20%
Average	20%	25%	35%	20%

Source: ChatGPT 5.2, Claude Opus 4.6, Deutsche Bank Research

There is general agreement on the likely impact of AI on inflation over the longer (5 year) timeframe. ChatGPT and Claude on balance see somewhat greater probabilities of lower inflation outcomes. Claude sees slightly reducing inflation as the most likely outcome (35%), which along with meaningfully reduces (20%) suggests a 35pp skew towards lower versus higher inflation. ChatGPT shows an even more notable skew towards lower inflation, with 60% applied to this outcome (40% for slight decline and 20% for meaningful decline) versus 15% for higher inflation scenarios. Across the tools, AI sees 50-60% probability of lower inflation over the 5-year horizon versus 15-25% probability it will lift inflation over that horizon. That said, across all the tools a positive impact on inflation is seen as roughly as likely as a meaningful reduction in inflation. Indeed, the average probability of each outcome is 20% across the tools.

We provide detailed reasoning in an appendix.

Conclusion

AI does not necessarily agree with the strong consensus that AI is likely to exert a meaningful drag on inflation pressures over the coming years. How should we interpret this finding?

One option is to discount the result. Surely AI must be disinflationary, the consensus believes, so AI is simply missing something. While perhaps appealing, it strikes us that this approach is fundamentally flawed — if AI is categorically wrong about this question, perhaps we should rethink our assessment of how transformative it is likely to be for complex knowledge work like forecasting, at least in its current form.

A second (and opposing) possibility is that the consensus is somewhat off base in assuming AI will be an unequivocal disinflationary force. Perhaps it is not accounting for the inevitable countervailing forces that will keep inflation



supported against the coming drag. Factors supporting demand, at least in the near term, are notable.

A middle ground is that AI is taking a sensible approach by assigning relatively flat probabilities across outcomes in a highly uncertain environment with longer time horizons. Having been trained on a corpus of text from economists, AI is simply acting as the proverbially two-handed economist, hedging its views against an unknowable backdrop.

As our colleague Jim Reid noted in his recent piece (see [“Innovation, jobs and inflation: Lessons from 250 years of disruption”](#)), the long history of inflation alongside transformative technologies suggests that some combination of the latter two explanations may be appropriate. History has shown that the force of the policy response will be at least as critical as the technology in determining the broader impact on inflation.

At this point, AI appears to largely agree.

Appendix

ChatGPT’s Reasoning:

1. *Why the 1-year horizon is centered on “minimal impact”*

Over a 1-year window, AI is unlikely to dominate US inflation dynamics. Inflation at that horizon is usually driven by monetary policy, wage growth, housing services, commodity prices, supply chains, and fiscal conditions. Even if AI adoption is rapid in some occupations, aggregate price indexes tend to move slowly because:

- General-purpose technologies often have delayed macro effects due to complementary investments and reorganization.
- Early adoption is concentrated in specific tasks and firms rather than economy-wide.
- Measured productivity gains often lag technical capability.
- Consumer price pass-through from lower business costs is partial and uneven.

This logic supports assigning the highest short-run probability to outcome B.

2. *Why 5 years shifts toward disinflation*

Over 5 years, it becomes more plausible that AI lowers inflation modestly. The main channels are:

- **Productivity:** AI can reduce labor input per unit of output in coding, customer support, back-office work, design, and analytics.
- **Logistics and allocation:** Better forecasting and optimization can lower inventories, waste, and delays.
- **Labor market effects:** If AI slows wage growth in exposed tasks or increases effective labor supply, unit labor costs may ease.



- **Competition:** Lower entry barriers and faster imitation may reduce pricing power in some sectors.

These channels support assigning the greatest 5-year mass to outcome C, with a meaningful tail probability on D.

3. *Why a strongly inflationary AI scenario remains possible*

AI could still raise inflation if demand effects exceed supply effects. For example:

- AI investment booms may raise demand for semiconductors, energy, cloud infrastructure, and specialized talent.
- Higher asset prices and stronger business confidence could stimulate aggregate demand.
- If AI increases market concentration or markups, firms may not fully pass cost savings to consumers.
- Adjustment costs may temporarily depress measured productivity.

This keeps a non-trivial probability on A at both horizons, though less so after 5 years.

4. *Why very large disinflation is not the base case*

A reduction of inflation by 0.5pp or more would require broad, fast, and pass-through-heavy productivity gains. That is possible, but not the central scenario, because:

- Diffusion across regulated and service-heavy sectors is often slow.
- Many gains may show up in quality improvements rather than lower measured prices.
- Some firms may retain gains as profits.
- Complementary bottlenecks, especially power and compute, could offset part of the supply benefit.

Claude's Reasoning:

1-Year Horizon (2025–2026)

At the 1-year horizon, the dominant expectation is that AI will have **minimal direct impact on aggregate US inflation** (Scenario B, 50%). The reasoning is as follows:

- **Adoption lags:** While ChatGPT and similar tools have achieved rapid user adoption, enterprise-level integration into production processes — the kind that affects unit costs and prices — takes considerably longer. Brynjolfsson, Rock, & Syverson (2021) document a "productivity J-curve" where new general-purpose technologies initially *reduce* measured productivity as firms invest in complementary intangible capital (reorganization, training, workflow redesign) before benefits emerge.
- **Small macro footprint:** AI-related spending, while growing rapidly, remains a small share of GDP. Total US AI investment (hardware, software, services) is estimated at roughly \$150–200 billion annually — significant but under 1% of GDP. Its price-level effects are correspondingly small relative to forces like monetary policy, fiscal policy, and global commodity markets.



- **Inflationary risk (Scenario A, 25%):** The probability of a slight inflationary impact is non-trivial because of the surge in AI-related capital expenditure. Data-center construction, semiconductor demand, and energy consumption for AI workloads are all increasing rapidly. The IEA (2024) projects data-center electricity consumption could double by 2026. This demand-pull inflation in energy and investment goods could slightly raise headline inflation, even if AI's net long-run effect is deflationary. Additionally, some firms are raising prices for AI-enhanced products and services.
- **Limited deflationary potential (Scenarios C & D, 20% & 5%):** Some early-adopting sectors (customer service, coding, content creation, basic legal/financial analysis) may already see modest cost reductions. However, these effects are too small and too narrowly distributed to materially affect the CPI within one year.

5-Year Horizon (2025–2030)

At the 5-year horizon, the probability distribution shifts meaningfully toward deflationary outcomes, with **Scenario C (slight reduction, 35%)** becoming the modal outcome. The reasoning:

- **Productivity gains materialize:** By 2030, AI will have had sufficient time for broad enterprise adoption. Based on historical GPT diffusion patterns (David, 1990; Brynjolfsson et al., 2021), 5–7 years is typically when measurable productivity gains begin appearing in macro data. Baily, Brynjolfsson, & Korinek (2023) project AI could add 0.4–0.9pp to annual labor productivity growth, which would translate to meaningful downward pressure on unit labor costs and thus prices.
- **Services inflation moderation:** Services constitute roughly 60% of the CPI basket and have been the primary driver of post-pandemic inflation persistence. AI is disproportionately suited to automating or augmenting service-sector tasks (Eloundou et al., 2023), suggesting it could address the stickiest component of inflation. Healthcare administration, financial services, legal services, education, and government services are all areas where AI could reduce costs.
- **Supply chain and logistics optimization:** AI-driven improvements in demand forecasting, inventory management, routing, and supply chain coordination could reduce goods-sector costs, compounding the services-side effects.
- **Scenario D tail risk (20%):** The probability of meaningful deflation (–0.5pp or more) is substantial at the 5-year horizon. This reflects the possibility of faster-than-expected AI capability improvements (e.g., toward more autonomous AI agents) and rapid adoption. If AI achieves the upper end of productivity estimates from Korinek & Suh (2024), the deflationary impact could be significant. However, Acemoglu (2024) provides a more skeptical baseline, arguing that AI's impact will be concentrated in a limited set of tasks and that general equilibrium effects (e.g., Baumol's cost disease in non-automated sectors) will limit aggregate impact.
- **Residual inflationary risk (Scenario A, 20%):** Even at the 5-year horizon, AI could prove inflationary if: (a) AI-driven productivity gains primarily accrue to capital owners, boosting wealth and demand without proportionate supply increases; (b) AI-driven energy demand significantly tightens electricity markets; (c) AI enables firms to engage in more effective price discrimination and dynamic pricing, extracting more consumer surplus; or (d) geopolitical competition over AI leads to industrial policy and subsidies that are inflationary.

Key Uncertainties and Caveats

- **Technology trajectory:** The pace of AI capability improvement is deeply uncertain. A slowdown in scaling laws would favor Scenarios A/B; breakthroughs in reasoning, planning, or autonomous agents would favor C/D.
- **Policy response:** The Federal Reserve will adjust monetary policy in response to AI's effects on inflation, potentially dampening both inflationary and deflationary extremes. Regulatory choices (AI safety requirements, licensing, labor protections) will also affect adoption speed and cost.
- **Pass-through dynamics:** Even if AI reduces production costs, the extent to which these savings are passed through to consumers (vs. captured as higher margins) depends on competitive structure. Highly concentrated industries may retain savings.
- **Labor market adjustment:** If AI displaces workers faster than new jobs are created, reduced consumer spending could create deflationary demand-side effects beyond the supply-side productivity channel.



Conversely, if labor markets remain tight due to demographic trends, AI-driven productivity gains may be necessary just to prevent inflation from rising.

- **These probabilities represent informed subjective estimates** based on available research as of mid-2025. They are not derived from a formal probabilistic model and carry substantial uncertainty.



Appendix 1

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